

Worksheet — Bandwidth, Download Time & Buffering

Independent tasks. Answer every question on this sheet. Use the Lesson 2 web page to help you.

Name: _____

Class: _____

Date: _____

Instructions for students

- Open the Lesson 2 web page: *Networks Year 9 → Lesson 2: Connectivity*.
- Start with the short **Recap**. Most of this worksheet is about **bandwidth**, **download time** and **buffering**, which are new today.
- Work on your own, in silence. Write full sentences where the question asks you to *explain* or *describe*.
- **Show your working for every calculation.** A correct answer with no working only earns half marks.
- If you finish early, complete the **Stretch challenge** at the bottom.

Recap

Last lesson: cables, submarine cables & Wi-Fi

/ 6 marks

Quick recall — do this from memory first, then check the Lesson 2 page if you need to.

1. Name the **three types of cable** covered last lesson.

2. Which cable type is the **fastest**, and what does it use to carry data?

3. What is a **submarine cable** and what is it used for?

4. Give **one advantage** and **one disadvantage** of using **Wi-Fi** instead of a wired connection.

Task 1 Bandwidth

/ 6 marks

Read the *Bandwidth* section on the Lesson 2 page.

1. Write a one-sentence definition of **bandwidth** in your own words.

2. The lesson uses a **water pipe** and a **motorway** to explain bandwidth. Pick one and explain how it helps you picture what bandwidth means.

3. What is the difference between **Mb/s** and **MB/s**? How many bits are in one byte?

4. A modem in the year 2000 ran at 56 kb/s. Modern home internet runs at around 100 Mb/s. Roughly how many times faster is modern home internet?

Task 2 Calculating download time

/ 10 marks

Use the *Calculating download time* section. Show all working — mark up each answer with what you divided and what unit you converted.

Unit reminder:

8 bits (b) = 1 byte (B)

1,000 bytes = 1 kilobyte (kB)

1,000,000 bytes = 1 megabyte (MB)

1,000,000,000 bytes = 1 gigabyte (GB)

To convert Mb/s → MB/s, **divide by 8**.

Method (from the worked example):

1. **Step 1** — Convert the bandwidth from bits/s into *bytes/s* ($\div 8$).
2. **Step 2** — Divide the file size by that speed. Answer = time in seconds.
3. Check the units match before you divide (both must be in bytes, or both in bits).

Q1. A home has a bandwidth of **80 Mb/s**. How long will it take to download a **200 MB** file?

Working:

Answer: _____ seconds

Q2. A café Wi-Fi runs at **40 Mb/s**. How long will a **2 GB** film take to download? Give your answer in *minutes and seconds*.

Working:

Answer: _____

Q3. Two friends both want to download a **4 GB** game. Ali has a **50 Mb/s** connection. Bea has a **200 Mb/s** connection. Calculate how long each of them waits, and work out how much time Bea saves compared with Ali. Give your answers in *minutes and seconds*.

Working:

Ali: _____

Bea: _____

Time saved: _____

Task 3 Buffering

/ 10 marks

Read the *Buffering* section and try both the **fast internet** and **slow internet** demos on the page before answering.

1. What is a **buffer**? Write a one-sentence definition.

2. Explain, in your own words, **why the video pauses to buffer** when the internet is slow. Use the words *bandwidth*, *buffer* and *playback* in your answer.

3. Why does a **faster bandwidth** mean fewer pauses when streaming?

4. A streaming service says a film needs **25 Mb/s** to play in 4K without buffering. A student's home internet runs at **15 Mb/s**. Describe what will happen when the film plays, and suggest **two** things the student could change to fix it.

5. Why does a video sometimes start playing **almost instantly** but stop a few seconds later? Link your answer to how the buffer fills up.

Task 4 **Key terms**

/ 4 marks

Use the *Plenary* key terms cards. Write a short definition for each.

Term	Definition
Bandwidth	
Buffer	
Mb/s vs MB/s	
Download time	

★ Stretch challenge (extension)

Finished everything? Try these — they need you to think, not just read.

1. Bandwidth is often advertised in Mb/s but file sizes are given in MB. Why do you think internet companies advertise using **Mb/s** and not MB/s? Is this fair on the customer?

2. A live football match streams in HD at around **8 Mb/s**. If 20 people in the same house all try to watch on different devices, the router has a total bandwidth of **100 Mb/s**. Explain what will happen, using the ideas of bandwidth and buffering.

3. Design a rule you could give a family to **reduce buffering** at peak times, without upgrading their internet package. Explain *why* your rule would work.
